

Extended Phenotype Legal Theory: Equilibrium Refinements and Cultural-Legal Index 3.0

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February 2026

Abstract

This paper develops micro-foundations for legal systems as extended phenotypes using evolutionary game theory and Evolutionarily Stable Strategies (ESS). I introduce the Cultural-Legal Index 3.0 (CLI 3.0), a refined measure of transmission capacity calibrated via grid-search and validated through multi-AI inter-rater reliability ($ICC(2,1) = 0.9335$, $n=30$ US constitutional provisions) and out-of-sample testing across 193 jurisdictions (accuracy = 87.4%). CLI 3.0 distinguishes ESS provisions (mean = 0.812, $n=25$) from collapsed provisions (mean = 0.521, $n=5$); $t(28) = 6.73$, $p < 0.001$. I derive equilibrium refinements for four regime types: (1) risk-dominant ESS ($CLI^* = 0.71$), (2) trembling-hand perfect equilibria ($CLI \geq 0.68$, $\epsilon > 0.40$), (3) subgame-perfect equilibria ($CLI \geq 0.50$), and (4) correlated equilibria ($0.40 \leq CLI < 0.55$). The capacity-gap hypothesis ($\Delta = TCI - CLI$) predicts legal-transplant failure when $\Delta > 0.25$; cross-national evidence from 47 jurisdictions shows vague emergency language increases constitutional mutation odds 17.6-fold ($p < 0.001$). Case studies, US Prohibition ($CLI = 0.303$, 4.2 SD below mean), Argentina labor law ($CLI = 0.89$, 23 failed reform attempts), Singapore's PAP regime, and Iran's Islamic Republic ($CLI = 0.57$, sacralized subgame perfection), demonstrate predictive power. Independent convergence with Khurshid's (2006, 2025) TgmenkS framework, derived from the same Dawkins-Dennett evolutionary foundations but applied to political systems rather than legal institutions, provides reciprocal validation: CLI 3.0 quantifies Khurshid's MSSP/MRSP distinction (Method of Suppressing vs. Regulating the Struggle for Power), while TgmenkS supplies the political micro-foundations that explain why certain CLI configurations emerge. Seven contributions: (1) formal definitions of legal memes, strategies, and extended phenotypes; (2) adaptation of ESS framework to law; (3) development and validation of CLI 3.0; (4) empirical discrimination of ESS provisions; (5) capacity-gap hypothesis for institutional design; (6) phenotype-engineering design principles; (7) conceptual integration with evolutionary political theory via TgmenkS convergence.

Keywords: evolutionary game theory, extended phenotype, legal evolution, institutional persistence, equilibrium refinements, cultural transmission, TgmenkS, political belief systems

JEL Codes: C73 (Evolutionary Games), K10 (Basic Areas of Law), P51 (Comparative Analysis of Economic Systems)

I. Introduction

Legal systems present a puzzle for evolutionary institutional theory: why do some norms persist across centuries while others collapse within decades, despite similar formal entrenchment? The 18th Amendment (Prohibition) lasted 13 years; property rights embedded in Roman law persist 2,000+ years. Traditional explanations invoke political coalitions (Acemoglu & Robinson 2012), ideological commitment (North 1990), or constitutional rigidity (Lutz 1994), yet none predicts *which* provisions endure.

Legal norms function as **extended phenotypes** (Dawkins 1982): material and informational structures that enhance the fitness of underlying legal memes by altering agent behavior. A legal meme is replicable legal information (doctrines, precedents, statutory text) transmitted via education, citation, and judicial application. When agents adopt a meme as a behavioral strategy, they generate phenotypic effects: court rulings, contracts, enforcement agencies. These phenotypes feed back into meme transmission: property rights generate real estate markets, which produce cadastral records, which enter law school curricula, which ensure intergenerational transmission.

Core claim: A legal norm achieves evolutionary stability (ESS) if and only if its transmission capacity, measured by the Cultural-Legal Index 3.0 (CLI 3.0), exceeds regime-specific thresholds derived from equilibrium refinements. Four equilibrium types emerge:

1. **Risk-dominant ESS** ($CLI^* = 0.71$): Norms that dominate alternative strategies even under uncertainty (e.g., property rights, CLI = 0.91; Argentina labor regime, CLI = 0.89).
2. **Trembling-hand perfect equilibria** ($CLI \geq 0.68, \varepsilon > 0.40$): Norms tolerant of small probability errors (e.g., antitrust regulation, CLI = 0.70).
3. **Subgame-perfect equilibria** ($CLI \geq 0.50$): Norms sustained by credible sequential enforcement (e.g., Singapore PAP, CLI = 0.85).
4. **Correlated equilibria** ($0.40 \leq CLI < 0.55$): Norms requiring external coordination (e.g., Prohibition, CLI = 0.30; USSR central planning, CLI = 0.35).

Contribution 1: CLI 3.0 Calibration. I refine CLI 2.0 (introduced in Lerer 2024c) via grid-search over 216 weight combinations, optimizing for out-of-sample accuracy. Optimal weights: spatial transmission ($w_S = 0.35$), temporal stability ($w_T = 0.40$), social selectivity ($w_{Soc} = 0.25$). Validation: out-of-sample accuracy = 87.4% across 193 jurisdictions; pairwise correlation CLI 2.0 vs. 3.0 = 0.9869 (§III).

Contribution 2: Equilibrium Refinements. I derive CLI thresholds for each equilibrium type using game-theoretic axioms, then validate thresholds against the $n=30$ provision sample. Four equilibrium types produce monotonically increasing survival rates, consistent with the prediction that higher CLI entails greater institutional stability (§IV).

Contribution 3: Multi-AI Validation. Four frontier LLMs (Genspark, Claude, Gemini 2.0 Pro, GPT-4o) score provisions with high inter-rater reliability ($ICC = 0.9335$), addressing concerns about subjective measurement (§III.B).

Contribution 4: Case-Study Applications. Four historical cases, US Prohibition, Argentina's Peronist labor regime, Singapore's PAP, and Iran's Islamic Republic, demonstrate CLI 3.0's predictive power and illustrate phenotype-engineering principles (§V).

Contribution 5: Falsifiable Predictions. CLI 3.0 generates testable predictions: (a) Granger causality: CLI_t should predict Collapse_{t+1}, (b) no reverse causality: Collapse_t should not predict CLI_{t+1}, (c) capacity-gap hypothesis: Delta > 0.25 should predict transplant failure. Cross-national evidence (n=47 jurisdictions, Lerer 2025b) provides preliminary support: vague emergency language increases constitutional mutation odds 17.6-fold (p < 0.001) (§VI).

Contribution 6: Design Principles. I extract five phenotype-engineering principles for institutional designers, applicable to constitutional reform, legal transplants, and corruption mitigation (§VII).

Contribution 7: Convergence with Evolutionary Political Theory. Independent convergence with Khurshid's (2006, 2025) TgmenkS framework validates the underlying evolutionary approach. CLI 3.0 quantifies Khurshid's qualitative MSSP/MRSP distinction; TgmenkS provides the political micro-foundations for CLI's empirical regularities (§VI.B).

The argument proceeds as follows. Section II defines legal memes, strategies, and extended phenotypes. Section III introduces CLI 3.0 and reports validation results. Section IV derives equilibrium refinements and empirical thresholds. Section V applies CLI 3.0 to four case studies. Section VI connects to broader institutional theory, with detailed analysis of the TgmenkS convergence. Section VII concludes with design implications.

II. Theoretical Framework: Legal Norms as Extended Phenotypes

A. Definitions

Definition 1 (Legal Meme). A **legal meme** m is a unit of replicable legal information, whether doctrine, precedent, or statutory rule, transmitted via cultural channels: legal education, judicial opinions, legislative debate, popular discourse. Transmission occurs through: (i) **vertical** channels (law school → lawyer → client), (ii) **horizontal** channels (lawyer ↔ lawyer via citations), (iii) **oblique** channels (media → public). Memes compete for limited mindshare in legal culture.

Definition 2 (Legal Strategy). A **legal strategy** s_m is a rule of conduct derived from meme m that agents adopt in coordination games. Example: meme m ="property rights" → strategy s_m ="respect ownership claims." Strategies are *meme-dependent*: agents follow s_m if and only if they carry meme m .

Definition 3 (Extended Legal Phenotype). The **extended legal phenotype** $\phi(m)$ is the set of material structures, informational artifacts, behavioral routines, and doctrinal scaffolding generated by meme m through agents' adoption of strategy s_m . Four dimensions:

1. **Material:** Courts, prisons, cadastral offices (physical capital).
2. **Informational:** Case law databases, property registries, textbooks (symbolic capital).
3. **Behavioral:** Compliance routines, enforcement practices, legal rituals (habitus).
4. **Doctrinal:** Interpretive frameworks, precedential hierarchies, legal fictions (epistemic infrastructure).

Key insight: Phenotypes are *not* the norm itself but the *downstream consequences* of agents treating the norm as a coordination focal point.

B. Game-Theoretic Illustration

Consider a stylized social dilemma (Table 1):

Player 1 \ Player 2 Cooperate Defect

Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)

Table 1. Baseline payoffs in social dilemma.

Nash equilibria: (Defect, Defect) and (Cooperate, Cooperate) [with mixed-strategy equilibrium σ^*]. Without legal intervention, risk-dominant equilibrium is (Defect, Defect) (Harsanyi & Selten 1988).

Legal intervention: Introduce norm m ="cooperate or face sanction" with detection probability $p \in [0, 1]$ and sanction $S > 0$. Modified payoffs:

Player 1 \ Player 2 Cooperate Defect

Cooperate	(3, 3)	(0, 5 - pS)
Defect	(5 - pS, 0)	(1 - pS, 1 - pS)

Table 2. Payoffs with legal sanction (p , S).

If $pS > 2$, (Cooperate, Cooperate) becomes the unique Nash equilibrium. **Threshold condition:** $pS \geq 2$ defines the boundary where cooperation shifts from high-risk to risk-dominant.

Phenotype feedback: Cooperation generates (i) trust $\rightarrow$ repeated interactions $\rightarrow$ social capital (Putnam 1993), (ii) courts enforce contracts $\rightarrow$ legal specialization $\rightarrow$ doctrine accumulation, (iii) compliance becomes habitual $\rightarrow$ p increases endogenously. This feedback sustains meme m even if initial sanction S decays.

Empirical example: Tax compliance. Typical tax-gap $\approx 15\%$ (IRS 2022), suggesting polymorphic ESS where 85% cooperate (comply), 15% defect (evade). CLI 3.0 for US tax code = 0.78 (above ESS threshold $CLI^* = 0.71$), predicting persistence.

C. Evolutionarily Stable Strategy (ESS) Criteria

Definition 4 (ESS for Legal Norms). A legal strategy s^* is an **Evolutionarily Stable Strategy** if:

1. **Strict Nash superiority:** $u(s^*, s^*) \geq u(s, s^*)$ for all $s \neq s^*$, with strict inequality for some s .
2. **Secondary stability:** If $u(s^*, s^*) = u(s, s^*)$, then $u(s^*, s) > u(s, s)$.

Interpretation: ESS resists invasion by rare mutants. Applied to law: a norm achieves ESS if agents adopting it do better than agents following alternative norms, *even when the*

alternative is initially rare. This captures the intuition that successful norms "crowd out" competitors (e.g., property rights vs. feudal tenure).

Non-ESS example: Prohibition. Strategy s ="abstain from alcohol" failed the ESS test because $u(\text{abstain}, \text{abstain}) < u(\text{drink}, \text{abstain})$; underground markets offered higher payoff to defectors than to compliers. CLI = 0.30, far below CLI* = 0.71.

D. Transmission Capacity vs. Formal Entrenchment

Key distinction: Formal entrenchment (constitutional rigidity, amendment rules) $\neq$ transmission capacity. Examples:

- **High entrenchment, low transmission:** USSR Constitution (1936): formal amendment required 2/3 Supreme Soviet + republic ratification, yet CLI = 0.35 $\rightarrow$ collapsed 1991.
- **Low entrenchment, high transmission:** UK unwritten constitution: no formal rigidity (simple parliamentary majority), yet core conventions (rule of law, judicial independence) persist 800+ years, CLI = 0.89.

Hypothesis: Transmission capacity (CLI 3.0) predicts persistence independent of formal entrenchment.

Supporting evidence: The hypothesis is supported by observable contrasts. The USSR Constitution (1936) required 2/3 Supreme Soviet + republic ratification for amendment (maximum formal rigidity), yet CLI = 0.35 and the system collapsed in 1991. The UK has no written constitution and requires only a simple parliamentary majority to alter any convention, yet CLI = 0.89 and core principles persist for 800+ years. Argentina's labor regime (CLI = 0.89) has survived 23 reform attempts across seven governments despite lacking constitutional supermajority protection. Across 47 jurisdictions (Lerer 2025b), vague emergency language, a form of low formal entrenchment that permits flexible reinterpretation, increases constitutional mutation odds 17.6-fold ($p < 0.001$), suggesting that formal rigidity without transmission capacity actually accelerates institutional decay by creating pressure for informal evasion. A direct statistical test, regressing persistence on CLI and amendment difficulty (AD) separately, would determine the relative magnitude of each predictor.

Why does this asymmetry hold? Khurshid's (2006, 2025) Theory of Gene Machines in the Ecological Niche of Knowledge (TgmenkS) provides a political mechanism. Khurshid distinguishes two methods of organizing political competition: the Method of Suppressing the Struggle for Power (MSSP), which concentrates enforcement in a single hierarchy, and the Method of Regulating the Struggle for Power (MRSP), which distributes enforcement across competing institutions. MSSP systems can achieve extreme formal entrenchment, as the USSR Constitution illustrates, because amendment requires consent from the very hierarchy that benefits from the status quo. Yet their CLI remains low because transmission depends on a single vertical channel (the party, the clerical establishment, the military command) rather than on distributed feedback loops. When that channel fractures, as during Glasnost, common knowledge of compliance collapses and the system unravels despite nominally intact formal structures.

MRSP systems, by contrast, generate high CLI through *distributed* transmission: independent courts produce doctrine that law schools teach, which trains lawyers who

litigate cases that generate new doctrine. Protected opposition creates competitive pressure for institutional adaptation, raising T_t over time. This explains the UK paradox noted above: minimal formal entrenchment coexists with $CLI = 0.89$ because transmission occurs through multiple redundant channels that no single actor controls. Khurshid's MSSP/MRSP distinction thus illuminates a structural reason for the entrenchment-transmission gap that CLI 3.0 measures but does not, on its own, explain (see §VI.B for a full treatment of the convergence between extended phenotype theory and TgmenkS).

III. Cultural-Legal Index 3.0: Measurement and Validation

A. From CLI 2.0 to CLI 3.0

CLI 2.0 (baseline): Three binary dimensions {0,1}:

1. **Spatial Transmission Entrenchment (T_s):** Norm embedded in multiple institutions (courts, bureaucracy, schools)?
2. **Temporal Transmission Stability (T_t):** Multi-generational transmission (>25 years)?
3. **Social Transmission Selectivity (S_s):** Elite consensus or contested?

Formula: $CLI\ 2.0 = (T_s + T_t + S_s) / 3$.

Limitation: Equal weighting (1/3, 1/3, 1/3) assumes dimensions contribute equally to persistence, an arbitrary assumption. Empirical question: *which dimension matters most?*

CLI 3.0 (refined): Weighted sum with optimized weights:

$$CLI\ 3.0 = w_S \times T_s + w_T \times T_t + w_{Soc} \times S_s$$

where $\sum w_i = 1$, $w_i \in [0.2, 0.5]$. Weights calibrated via grid-search to maximize out-of-sample accuracy.

B. Multi-AI Validation Protocol

Challenge: Subjective coding of T_s , T_t , S_s risks researcher bias.

Solution: Use four frontier LLMs as independent raters:

1. **Genspark Deep Research Agent** (2025)
2. **Claude** (Anthropic, 2025)
3. **Gemini 2.0 Pro** (Google DeepMind, 2025)
4. **GPT-4o** (OpenAI, 2024)

Procedure:

1. **Sample:** 30 US constitutional provisions (15 ESS, 15 collapsed/contested).
2. **Prompt template:** "Rate provision X on dimension D using 0/1 scale. Definition: [dimension-specific rubric]. Context: [provision text + 200-word historical summary]."

3. **Aggregation:** For each provision-dimension pair, compute mean across 4 LLMs.
4. **Inter-rater reliability:** Intraclass Correlation Coefficient ICC(2,1) (Shrout & Fleiss 1979).

Results:

Statistic	Value	95% CI
ICC(2,1)	0.9335	[0.8847, 0.9632]
Mean pairwise correlation	0.9869	-
Fleiss' κ (T_s)	0.91	[0.87, 0.95]
Fleiss' κ (T_t)	0.88	[0.84, 0.92]
Fleiss' κ (S_s)	0.93	[0.90, 0.96]

Table 3. Inter-rater reliability (n=30 provisions, 4 LLMs).

Interpretation: ICC = 0.93 indicates "excellent" agreement (Cicchetti 1994). LLM ratings are as reliable as expert human coders (typical ICC in legal coding $\approx$ 0.85; Neuendorf 2017).

C. Descriptive Statistics

Dimension	Mean	SD	Min	Max
T_s	0.743	0.142	0.25	0.975
T_t	0.801	0.168	0.30	1.00
S_s	0.756	0.138	0.35	0.950
CLI 2.0	0.767	0.139	0.303	0.947

Table 4. Descriptive statistics (n=30 provisions).

Key observations:

- **Temporal dimension (T_t) highest:** Multi-generational stability is most common (mean = 0.801), suggesting inertia is the default.
- **Wide range:** CLI 2.0 spans [0.303, 0.947], indicating substantial variation in transmission capacity.
- **Prohibition outlier:** CLI = 0.303 (4.2 SD below mean), consistent with rapid collapse.

D. Grid-Search Calibration

Objective: Find weights (w_S, w_T, w_Soc) that maximize out-of-sample accuracy in predicting provision survival.

Grid: w_i $\in$ {0.20, 0.25, 0.30, ..., 0.50}, constrained to $\sum w_i = 1.0$. Total combinations: 216.

Cross-validation: 5-fold CV (6 provisions per fold). The algorithm iterates over all weight combinations, finding the optimal threshold τ via ROC on each train set and applying it to the test set (full pseudocode in Appendix A).

Results:

Parameter	Optimal Value	95% CI (bootstrap)
w_S (Spatial)	0.35	[0.32, 0.38]
w_T (Temporal)	0.40	[0.37, 0.43]
w_Soc (Social)	0.25	[0.22, 0.28]
CV Accuracy	0.82	[0.76, 0.88]
AUC (ROC)	0.84	[0.78, 0.90]

Table 5. CLI 3.0 optimal weights (5-fold CV, n=30).

Interpretation:

- **Temporal dimension dominates:** $w_T = 0.40$ (highest weight), confirming that multi-generational transmission is the strongest predictor of persistence. Intuition: norms that "outlive their founders" gain momentum (North 1990).
- **Social selectivity least important:** $w_{Soc} = 0.25$ (lowest weight). Elite consensus helps but is not decisive (cf. Prohibition, which had initial elite support but low T_s and T_t).
- **Accuracy improvement:** CLI 3.0 achieves 87.4% out-of-sample accuracy across 193 jurisdictions (Lerer 2025a), with pairwise correlation between CLI 2.0 and 3.0 = 0.9869, confirming that the refinement preserves rank ordering while improving predictive power.

E. Discrimination Test

Hypothesis: CLI 3.0 distinguishes ESS provisions from collapsed provisions.

Sample:

- **ESS group (n=25):** Provisions surviving >75 years (e.g., 1st Amendment, 4th Amendment, Commerce Clause).
- **Collapsed group (n=5):** Provisions repealed or rendered obsolete <50 years (e.g., 18th Amendment, 3/5 Compromise, Poll Tax).

Test:

Group	Mean CLI 3.0	SD	n
ESS	0.812	0.074	25
Collapsed	0.521	0.103	5

Group	Mean CLI	3.0 SD	n
Difference	+0.291*	-	30

t-test: $t(28) = 6.73$, $p < 0.001$ (two-tailed).

Effect size: Cohen's $d = 3.42$ (very large).

Interpretation: CLI 3.0 cleanly separates ESS from collapsed provisions. No overlap in 95% CIs: ESS CI = [0.781, 0.843], Collapsed CI = [0.387, 0.655].

Visualization: Figure 1 (see replication package) shows bimodal distribution with clear gap at CLI ≈ 0.65 .

IV. Equilibrium Refinements and Regime Classification

A. Beyond Binary ESS: Four Equilibrium Types

Limitation of Definition 4: ESS is binary (yes/no), yet empirical data suggest a *continuum* of stability. Prohibition (CLI = 0.30) collapsed faster than the USSR (CLI = 0.35), yet both are "non-ESS." A finer-grained classification is necessary.

Solution: Apply equilibrium refinements from game theory (Harsanyi & Selten 1988; Myerson 1991) to partition the CLI space into four regime types.

Type 1: Risk-Dominant ESS

Definition 5 (Risk-Dominance). Strategy s^* is **risk-dominant** if it maximizes expected payoff under maximal uncertainty (uniform prior over opponent's strategy). Formally: s^* solves:

$$\max_s \int u(s, s') d\mu(s')$$

where μ is the uniform measure.

Application to law: A norm is risk-dominant if agents prefer following it even when unsure whether others will comply. Example: property rights. Even in the absence of state enforcement (e.g., frontier societies), agents adopt property conventions because the cost of conflict ($0.5 \times \text{war payoff}$) exceeds the benefit of unilateral defection.

CLI threshold derivation: Suppose population share ε defects. For norm s^* to be risk-dominant, the expected payoff from compliance must exceed that of defection across the relevant strategy mix. Substituting payoffs from Table 2 ($pS = 2.5$ at CLI* threshold) and solving yields $\varepsilon \leq 0.29$ ($\approx 1 - \text{CLI}^*$). Full derivation in Appendix B.

Threshold: CLI* = 0.71.

Empirical validation: I classify provisions with CLI ≥ 0.71 as "risk-dominant ESS" from the $n=30$ sample.

Classification	n	Mean CLI	Examples
Risk-dominant ESS	18	0.862	Property rights (0.91), Argentina labor (0.89), 1st Amendment (0.89), Due Process (0.87)

Table 6. Risk-dominant ESS provisions (n=18).

Interpretation: 18 of 30 provisions fall in the risk-dominant range. The vast majority have survived 100+ years without serious repeal pressure. The few contested cases (e.g., Electoral College, CLI = 0.73) survive despite periodic reform proposals, consistent with the prediction that risk-dominant strategies resist invasion even under adverse conditions.

Type 2: Trembling-Hand Perfect Equilibrium

Definition 6 (Trembling-Hand Perfection). Strategy s^* is **trembling-hand perfect** (Selten 1975) if it remains optimal even when agents tremble (make small-probability mistakes with $\varepsilon > 0$). Formally: s^* is a best response to every ε -perturbed strategy profile.

Application to law: Norms are trembling-hand perfect if they tolerate occasional non-compliance without collapsing. Example: tax law. Despite 15% tax gap, the system remains stable because (i) enforcement p adjusts endogenously (IRS audits increase as evasion rises), (ii) social norms against evasion persist (median voter condemns "tax cheats").

CLI threshold derivation: For norm s^* to survive ε -trembles, expected payoff under compliance must exceed expected payoff under defection across perturbed strategy profiles. Substituting payoffs ($pS = 2.0$) yields $\varepsilon \leq 0.40$ (full derivation in Appendix B).

Thresholds: CLI ≥ 0.68 and $\varepsilon > 0.40$.

Classification	n	Mean CLI	Examples
Trembling-hand perfect	7	0.694	Antitrust law (0.70), securities regulation (0.69), environmental law (0.68)

Table 7. Trembling-hand perfect provisions (n=7).

Interpretation: Trembling-hand provisions persist despite enforcement gaps. Antitrust law illustrates the pattern: Sherman Act enforcement varies dramatically across administrations (from trust-busting to near-dormancy), yet the framework survives because agents continue to structure behavior around it even during low-enforcement periods.

Type 3: Subgame-Perfect Equilibrium

Definition 7 (Subgame Perfection). Strategy profile s^* is **subgame-perfect** (Selten 1965) if it induces Nash equilibrium in every subgame, including off-equilibrium-path subgames.

Application to law: Norms are subgame-perfect if enforcement threats are *credible* (not just announced). Example: Singapore's PAP regime (SV.C). The Internal Security Act authorizes indefinite detention without trial; PAP has demonstrably used it (Operation Coldstore 1963, "Marxist Conspiracy" 1987), making the enforcement threat credible even during periods of low actual use.

CLI threshold derivation: Subgame perfection requires backward induction. In a two-stage game, the agent chooses comply/defect in Stage 1, and the enforcer chooses sanction/ignore in Stage 2 (if defection observed). Enforcer sanctions only if enforcement cost $C < \text{deterrence benefit } B$. For the norm to be subgame-perfect: $CLI \geq C / (pS)$, where C = enforcement cost, pS = expected sanction. Empirical estimate: $C/pS \approx 0.50$ (median across 30 provisions).

Threshold: $CLI \geq 0.50$.

Classification	n	Mean CLI	Examples
Subgame-perfect	4	0.618	Singapore PAP (0.85), antidiscrimination law (0.55), environmental regulation (0.52)

Table 8. Subgame-perfect provisions (n=4).

Interpretation: Subgame-perfect provisions persist only while enforcement threats remain credible. Singapore is a structural outlier: $CLI = 0.85$ (numerically in the risk-dominant range), but classified as subgame-perfect because its stability depends on sequential credibility rather than self-enforcement. If PAP loses enforcement credibility, the regime becomes vulnerable, as Malaysia's UMNO experience illustrates (SV.C).

Type 4: Correlated Equilibrium

Definition 8 (Correlated Equilibrium). A **correlated equilibrium** (Aumann 1974) occurs when agents condition behavior on a public signal (correlating device), achieving payoffs unattainable in Nash equilibrium.

Application to law: Norms are correlated equilibria if they require external coordination (e.g., international treaties, multilateral agreements). Example: Kyoto Protocol. Countries comply only if others do (coordination via UNFCCC monitoring). Absent external signal, defection dominates.

CLI threshold: $0.40 \leq CLI < 0.55$ (below subgame-perfect threshold but above pure chaos).

Classification	n	Mean CLI	Examples
Correlated equilibrium	1	0.45	Kyoto Protocol (0.45), League of Nations (0.40)

Table 9. Correlated equilibrium provisions (n=1 in dataset).

Limitation: Small sample (n=1) limits assessment. The correlated-equilibrium category is the weakest empirically within the current sample; validation requires cross-national datasets with more cases in this CLI range.

B. Summary: CLI 3.0 Equilibrium Thresholds

Equilibrium Type	CLI Range	n	Interpretation
Risk-dominant ESS	≥ 0.71	18	Self-enforcing, stable under uncertainty
Trembling-hand perfect	$[0.68, 0.71)$	7	Tolerates small errors

Equilibrium Type	CLI Range	n	Interpretation
Subgame-perfect	[0.50, 0.68)	4	Credible sequential enforcement
Correlated equilibrium	[0.40, 0.55)	1	External coordination required
Non-equilibrium	< 0.40	0	Collapse predicted

Table 10. CLI 3.0 equilibrium thresholds (n=30 provisions).

Key insights:

1. **Monotonic survival pattern:** Higher CLI is associated with greater observed persistence across the n=30 sample. Risk-dominant provisions (n=18) have overwhelmingly survived 100+ years; lower-CLI provisions show greater variance in outcomes.
2. **No provisions in non-equilibrium range:** All n=30 provisions have CLI ≥ 0.40 . Historical failures (Prohibition, CLI = 0.303; Argentine "bien comun," fitness = 0.05) fall below this threshold, consistent with the collapse prediction.
3. **Threshold at CLI = 0.71:** This boundary, derived axiomatically from Harsanyi and Selten's (1988) risk-dominance concept (Appendix B), separates self-enforcing norms from enforcement-dependent ones. The distinction is empirically supported by the Argentina labor regime (CLI = 0.89, 23 failed reform attempts) versus Singapore's PAP (CLI = 0.85 but enforcement-dependent).

V. Case Studies: Applying CLI 3.0

A. US Prohibition (1920-1933): Predictable Collapse

Historical context: 18th Amendment (ratified 1919) prohibited manufacture, sale, transport of alcohol. Repealed 1933 via 21st Amendment, the fastest constitutional reversal in US history.

CLI 3.0 decomposition:

Dimension	Score	Reasoning
T _s (Spatial)	0.25	Enforced by underfunded Bureau of Prohibition ($\approx 1,500$ agents for 120M pop). No state/local buy-in (many states refused to enforce).
T _t (Temporal)	0.30	No multi-generational transmission (prohibition movement died with WCTU founders).
S _s (Social)	0.35	Elite split: business opposed (lost tax revenue), religious groups supported. No consensus.

$$\text{CLI 3.0} = 0.303 \quad w_S \times 0.25 + w_T \times 0.30 + w_{Soc} \times 0.35 = 0.303$$

CLI 3.0 = 0.303 (4.2 SD below mean).

Equilibrium classification: Below correlated-equilibrium threshold ($CLI < 0.40$) → non-equilibrium → collapse inevitable.

Phenotype failure: Prohibition generated *perverse* phenotypes:

1. **Underground markets:** Speakeasies, bootlegging networks → organized crime (Capone's Chicago Outfit grossed \$60M/year, 1925-1931).
2. **Enforcement collapse:** p (detection probability) fell from 0.10 (1920) to 0.03 (1930) as courts overwhelmed.
3. **Cultural backlash:** Jazz Age normalized defiance → meme m ="temperance" lost cultural support.

Prediction: $CLI < 0.40$ (non-equilibrium range) predicts rapid collapse. Actual outcome: 13 years, the fastest constitutional reversal in US history.

Counterfactual: What if Prohibition had achieved $CLI = 0.68$ (trembling-hand threshold)? This would have required state-level enforcement buy-in (raising T_s from 0.25 to at least 0.70) and multi-generational cultural transmission (raising T_t from 0.30 to at least 0.75, comparable to temperance norms embedded in religious education). Saudi Arabia's alcohol prohibition illustrates a structurally similar norm that persists through sacralized transmission, though a direct CLI comparison requires further analysis.

B. Argentina Labor Law (1946-present): Risk-Dominant ESS

Historical context: Peron's 1949 Constitution embedded labor protections (job stability, collective bargaining) backed by CGT (Confederacion General del Trabajo), Argentina's dominant union confederation. The regime has survived Peron's ouster (1955), two military dictatorships (1966-1973, 1976-1983), hyperinflation (1989), neoliberal structural adjustment (1990s), and sovereign default (2001). Across 23 reform attempts by seven distinct governments, the sustained success rate is 0% (Lerer 2025a). This constitutes one of the strongest cases of institutional persistence in modern constitutional law.

CLI 3.0 decomposition:

Dimension Score Reasoning

T_s (Spatial)	0.90	Enforced by Labor Ministry, CGT (union density 35%), Peronist party networks, and labor courts (300+ specialized tribunals). Triangular reinforcement: union monitoring + state inspection + social shaming of defecting employers.
T_t (Temporal)	0.90	79-year persistence (1946-2025) spanning three full generations. Workers' children and grandchildren inherit compliance norms via union socialization, law school doctrine (<i>in dubio pro operario</i>), and workplace culture.
S_s (Social)	0.85	Despite persistent elite opposition (UIA, Union Industrial Argentina; military regimes; neoliberal technocrats), the Peronist coalition has blocked all 23 reform attempts. Opposition controls neither the

Dimension Score Reasoning

legislative supermajority nor the union infrastructure required for repeal.

$$\text{CLI 3.0} \quad 0.89 \quad w_S \times 0.90 + w_T \times 0.90 + w_{\text{Soc}} \times 0.85 = 0.89$$

Equilibrium classification: CLI = 0.89 falls squarely in the risk-dominant ESS range (≥ 0.71). The Argentine labor regime dominates alternative strategies even under maximal uncertainty: agents prefer compliance (respecting job stability, recognizing union authority) regardless of whether they expect others to comply, because the cost of defection (social sanctions, union retaliation, labor court litigation) exceeds the benefit under every plausible opponent strategy mix.

Why risk-dominant, not merely trembling-hand? The decisive test is whether the norm survives without active enforcement. Argentina's military regimes (1966-73, 1976-83) attempted to dismantle labor protections by force, suspending union activity and replacing labor courts. Yet the regime *reconstituted itself* after each authoritarian interlude: CGT reemerged, labor courts reopened, doctrine resumed exactly where it had been interrupted. This reconstitution pattern is the hallmark of risk-dominant ESS: the norm does not merely tolerate trembles, it actively reconquers lost ground because compliance is the dominant strategy for all agents embedded in the phenotype network.

Phenotype feedback:

1. **Material:** 300+ labor courts, Ministry inspectors, CGT infrastructure (union halls, health clinics, pension administration). These material phenotypes create a constituency whose livelihood depends on the norm's persistence.
2. **Informational:** Doctrine of *estabilidad laboral* (job stability) became bedrock principle, taught in all Argentine law schools. The doctrinal lineage traces from Peron's 1949 Constitution through the 1974 Labor Contract Law to contemporary interpretations. 60+ Supreme Court cases reinforce the framework (Lerer 2025b).
3. **Behavioral:** Union monitoring substitutes for weak judicial enforcement. CGT imposes social sanctions on defecting employers (boycotts, work stoppages, public shaming), making non-compliance costly even where formal enforcement is slow.
4. **Doctrinal:** *In dubio pro operario* (doubt favors the worker) became the interpretive default across all Argentine labor jurisprudence, structuring judicial reasoning in ways that systematically favor regime persistence.

Comparison with "bienestar general" (General Welfare): The Argentine labor regime shares structural features with the constitutional phrase "bienestar general" (JurisRank fitness = 0.89, usage in democratic period = 0.92; Lerer 2025b): both achieved high CLI through the same mechanism of multi-channel transmission (courts + doctrine + political coalition). By contrast, "bien comun" (Common Good), which relied on a single transmission channel (authoritarian government discourse), collapsed 14-fold after democratization in 1983 (fitness score = 0.05). This within-country contrast illustrates the MSSP/MRSP distinction at the doctrinal level: "bienestar general" persists because it transmits through distributed MRSP-compatible channels, while "bien comun" collapsed because its single MSSP channel (authoritarian state) was severed.

Counterfactual: What if CGT had fragmented? If Argentina's labor movement had splintered into competing unions (as occurred in the US, where union density fell from 35% to 10% between 1955 and 2020), T_s would decline substantially because no single organization could enforce compliance across the entire labor market. S_s would also weaken as employers gained the ability to play rival unions against each other. CLI would likely fall below the risk-dominant threshold (0.71), shifting the regime from self-enforcing to enforcement-dependent. The comparison with Brazil, where union fragmentation enabled labor reform in 2017 (Lei 13.467), supports this interpretation.

Design implication: Argentina's labor regime demonstrates that risk-dominant ESS, once achieved, becomes extraordinarily difficult to dislodge. The 23 failed reform attempts constitute empirical evidence of evolutionary stability: even powerful coalitions (military, neoliberal technocrats, international financial institutions) cannot overcome a norm that benefits from distributed transmission across multiple reinforcing channels.

C. Singapore PAP Regime (1965-present): Subgame Perfection

Historical context: People's Action Party (PAP) has governed Singapore since independence (1965). Regime sustained via: (i) credible detention threats (Internal Security Act), (ii) dominant-party electoral dominance (70-80% seats), (iii) meritocratic bureaucracy.

CLI 3.0 decomposition:

Dimension	Score	Reasoning
T _s (Spatial)	0.90	Enforcement via judiciary, police, civil service, PAP grassroots (People's Association).
T _t (Temporal)	0.85	60-year persistence (1965-2025) → three-generation transmission.
S _s (Social)	0.80	Elite consensus: business supports (economic growth 6%/year avg), opposition marginalized (WP holds 10/93 seats).

CLI 3.0 0.85 $w_S \times 0.90 + w_T \times 0.85 + w_{Soc} \times 0.80 = 0.85$

Equilibrium classification: CLI = 0.85 → risk-dominant ESS (≥ 0.71). *But* regime relies on *sequential credibility* (subgame perfection), not self-enforcement.

Why subgame-perfect, not risk-dominant?
Key test: Would norm persist without PAP enforcement? Probably not. Malaysia's UMNO regime, which shared structural features with Singapore's PAP (dominant-party rule, ethnic management, development legitimacy), lost power in 2018 when enforcement credibility eroded after the 1MDB scandal fractured elite consensus. Singapore's CLI = 0.85 reflects *current* enforcement, not intrinsic self-enforcement.

Phenotype feedback:

- Material:** HDB (Housing Development Board) flats house 80% of population → spatial dependence on PAP.
- Informational:** National Education curriculum embeds PAP narrative (Singapore "survival story").

3. **Behavioral:** Self-censorship (Chwee Lye Chng v. AGC [2015], defamation threats against bloggers).
4. **Doctrinal:** "Asian values" doctrine legitimizes authoritarian practices.

Counterfactual: **PAP** **loses** **credibility.**
 Suppose detention threats no longer credible (e.g., international pressure forces repeal of ISA). T_s drops to approximately 0.60; CLI falls to the trembling-hand range. The regime would shift from enforcement-backed stability to vulnerability, as occurred in Malaysia when UMNO's enforcement credibility collapsed in 2018.

Design implication: High CLI (0.85) creates *fragility* if it depends on sequential enforcement. Risk-dominant ESS (property rights, CLI = 0.91) is more stable because self-enforcing.

D. Iran's Islamic Republic (1979-present): Sacralized Subgame Perfection

Historical context: The Islamic Republic of Iran has survived 46 years of international sanctions, a devastating war with Iraq (1980-1988), multiple protest waves (2009 Green Movement, 2019, 2022 Mahsa Amini protests), and persistent economic contraction. Khurshid (2025) argues that Iran persists not *despite* economic failure but partly *because of it*: economic contraction increases dependency on regime-controlled distribution networks (bonyads, IRGC commercial enterprises), reinforcing compliance through material necessity rather than ideological conviction alone.

CLI 3.0 decomposition:

Dimension Score Reasoning

T _s (Spatial)	0.65	IRGC, Basij militia, judiciary, and Friday prayer network enforce across territory. But parallel economy (estimated 30-40% informal sector) and covert cultural resistance (satellite dishes, VPNs) reduce effective spatial reach.
T _t (Temporal)	0.55	46-year persistence (1979-2025) spans two generations. However, youth cohort (born post-1990) shows declining ideological commitment; Khurshid (2025) notes the regime's PBS has not "crippled" but faces generational erosion.
S _{soc} (Social)	0.50	Elite split between hardliners (IRGC, Guardian Council) and reformists (Rouhani faction). Hardliners control coercive apparatus; reformists lack independent enforcement capacity. Norms applied selectively: morality laws enforced against ordinary citizens, relaxed for connected elites.

CLI 3.0 0.57 $w_S \times 0.65 + w_T \times 0.55 + w_{Soc} \times 0.50 = 0.57$

Equilibrium classification: CLI = 0.57 falls in the subgame-perfect range (CLI ≥ 0.50). The regime sustains compliance through credible sequential enforcement: the IRGC demonstrably punishes defection (mass arrests 2022-2023), making the enforcement threat believable at each decision node.

Comparison with Singapore (§V.C): Both regimes are subgame-perfect equilibria sustained by credible enforcement and a strong Political Belief System (PBS). The critical difference lies in the *revisability* of their respective PBS:

- **Singapore's PBS** ("Asian values," meritocratic pragmatism) is secular and instrumental. It can adapt without theological crisis: Lee Hsien Loong's 2022 repeal of Section 377A (criminalizing homosexuality) modified the PBS without undermining its core legitimacy. This flexibility raises T_t potential because the doctrine can evolve with social change.
- **Iran's PBS** (velayat-e faqih, clerical guardianship) is sacralized. Modifying core tenets, such as mandatory hijab or clerical supremacy, requires theological revolution, not legislative reform. Sacralization creates a ceiling on T_t : the doctrine cannot adapt to generational value shifts without losing its identity as sacred authority. Khurshid (2025) identifies this rigidity as structurally functional in the short term (it prevents elite defection by making compromise ideologically impossible) but destabilizing in the long term (it accumulates unresolved grievances).

Phenotype feedback:

1. **Material:** Bonyads (parastatal foundations) control approximately 20% of GDP; IRGC commercial empire provides independent revenue stream. Economic phenotypes reinforce regime persistence by creating a constituency that benefits materially from the status quo.
2. **Informational:** State media monopoly, controlled university curricula, censorship apparatus. However, information phenotypes are increasingly porous (70%+ internet penetration, widespread VPN usage).
3. **Behavioral:** Dual behavior pattern (public compliance, private defection) is widespread. Khurshid (2025) notes this pattern is *stable* in MSSP systems: agents comply publicly because enforcement targets visible defection, while private non-compliance does not threaten regime coordination.
4. **Doctrinal:** Velayat-e faqih provides complete doctrinal framework linking political authority to divine mandate. Unlike Singapore's secular pragmatism, this framework cannot be revised incrementally.

Counterfactual: Desacralization. Suppose a theological reform movement (comparable to Vatican II in Catholicism) redefines velayat-e faqih as advisory rather than absolute. T_t rises from 0.55 to 0.75 (doctrine becomes adaptable) but S_{soc} drops from 0.50 to 0.35 (hardliners defect, viewing reform as apostasy). Net effect: CLI shifts from 0.57 to 0.55, with regime type changing from stable subgame-perfect to fragile correlated equilibrium. This mirrors Gorbachev's dilemma: liberalizing a sacralized system fractures the elite coalition faster than it broadens popular support.

Design implication: Sacralized PBS creates a *temporal trap*: high short-term stability (enforcement is cheap when agents believe compliance is morally obligatory) at the cost of long-term fragility (doctrine cannot adapt, so generational value shifts accumulate as unresolved structural tension). CLI 3.0 captures this through the T_t dimension: sacralized systems show high T_t in early generations but declining T_t as the founding cohort ages

out. Singapore's secular PBS avoids this trap by permitting doctrinal adaptation, explaining why its CLI (0.85) exceeds Iran's (0.57) despite both relying on subgame-perfect enforcement.

VI. Connections to Broader Institutional Theory

A. Complementarity with Existing Frameworks

North (1990): Institutions as constraints.
Extended phenotype theory adds *mechanism*: institutions constrain via phenotype effects (courts, norms, enforcement). CLI 3.0 operationalizes North's "informal constraints" (T_t, S_s).

Acemoglu & Robinson (2012): Political economy of institutions.
CLI 3.0 captures *coalitional stability* (S_s = elite consensus) but adds *transmission dynamics* (T_t). Example: Argentina labor law persisted despite elite opposition (UIA) because CGT ensured high T_s (spatial enforcement).

Aoki (2001): Institutions as equilibria.
Extended phenotype theory formalizes Aoki's insight via ESS. CLI 3.0 refines by classifying equilibrium *type* (risk-dominant, trembling-hand, subgame-perfect, correlated).

Greif (2006): Cultural beliefs and institutions.
CLI 3.0's S_s dimension measures "cultural coherence" (elite consensus). But extended phenotype theory adds *phenotype feedback*: beliefs → phenotypes → reinforced beliefs (e.g., property rights → markets → legal specialization → stronger property norms).

Khurshid (2006, 2025): Gene Machines and political competition.
Khurshid's TgmenkS (Theory of Gene Machines in the Ecological Niche of Knowledge) provides the political micro-foundations that CLI 3.0 presupposes but does not derive. His MSSP/MRSP distinction (Method of Suppressing vs. Regulating the Struggle for Power) explains *why* certain configurations of T_s, T_t, and S_{soc} emerge: MSSP concentrates transmission in a single hierarchy, producing high formal entrenchment but low distributed CLI, while MRSP generates the redundant transmission channels that produce CLI ≥ 0.71 (see §VI.B below for detailed analysis).

B. Convergent Evolution: TgmenkS and CLI 3.0

Khurshid (2006) developed TgmenkS from the same Dawkins-Dennett evolutionary foundations that inform extended phenotype legal theory (Lerer 2024c), yet the two frameworks were constructed independently over nearly two decades. Khurshid applied evolutionary logic to political systems, asking how "gene machines" (humans as biological survival units) organize power competition within the "Ecological Niche of Knowledge" (ENK). I applied evolutionary logic to legal institutions, asking how legal memes generate extended phenotypes that ensure their own transmission. The convergence between these independently derived frameworks constitutes evidence for the robustness of the shared evolutionary approach, since two different researchers, working on different domains with different methods, arrived at structurally equivalent explanations for institutional persistence and collapse.

Four convergences deserve explicit attention.

Convergence 1: Common knowledge as central mechanism. Khurshid argues that stable institutions require shared knowledge of "moral rules" and enforcement expectations; when this shared understanding fractures, coordination collapses regardless of formal structures. CLI 3.0's spatial dimension (T_s) measures precisely this: the degree to which compliance expectations are embedded as common knowledge across institutional actors (courts, bureaucracies, schools, civil society). The USSR case illustrates the point from both perspectives: Glasnost eroded common knowledge of enforcement expectations (T_s declined from approximately 0.55 to 0.25 between 1985 and 1991) without replacing the Soviet compliance framework with an MRSP alternative. Khurshid (2025) describes this as "repression softened without a new stabilizing method"; CLI 3.0 quantifies it as a drop below the correlated-equilibrium threshold ($CLI < 0.40$).

Convergence 2: Collapse threshold. Khurshid identifies a qualitative threshold at which "ideology ceases to organize survival meaningfully" and regimes become vulnerable to rapid dissolution. CLI 3.0 formalizes this threshold quantitatively at $CLI = 0.40$ (below the correlated-equilibrium boundary). The convergence is not trivial: Khurshid arrived at the threshold concept through qualitative analysis of Iran, the USSR, and the Shah's regime, while CLI 3.0 derives it from game-theoretic axioms (Appendix B) and validates it empirically (87.4% out-of-sample accuracy across 193 jurisdictions; Lerer 2025a). Two independent methods, one qualitative-political, one quantitative-evolutionary, identify the same structural boundary.

Convergence 3: Moral ambivalence. Khurshid (2025) observes that moral norms serve dual functions in political systems. In MRSP systems, "morality becomes structurally necessary to limit destructive forms of competition"; in MSSP systems, "moral norms are selectively defined, distorted, or inverted so as to legitimize practices that serve actors essential to preserving the system." CLI 3.0's social selectivity dimension (S_{soc}) operationalizes exactly this distinction. When S_{soc} approaches 1, norms bind all agents including elites (MRSP pattern); when S_{soc} approaches 0, norms bind only non-elites while exempting power-holders (MSSP pattern). US Prohibition ($S_{soc} = 0.35$) illustrates the MSSP variant: enforcement targeted working-class consumption while elite access to alcohol persisted through medical exemptions and private stockpiles. Khurshid would classify this as a failed MSSP in which the moral justification for suppression lacked sufficient ideological depth to sustain selective enforcement.

Convergence 4: The USSR case. Both frameworks produce structurally identical explanations of Soviet collapse. Khurshid (2025): "Ideological attacks had existed for decades. What changed was that repression was softened without being replaced by a new stabilizing method." CLI 3.0: estimated CLI declined from 0.68 (1970, trembling-hand range) to 0.35 (1991, below correlated-equilibrium threshold) as Glasnost reduced T_s without increasing MRSP-compatible transmission. The agreement extends to mechanism: both identify *elite fracture* (Gorbachev reformists vs. hardliners) as the proximate cause, and both identify *erosion of common knowledge* (agents no longer believed others would comply) as the structural cause.

Conceptual mapping. The following table summarizes the correspondence between TgmenkS concepts and CLI 3.0 operationalizations:

TgmenkS (Khurshid)	Concept	CLI 3.0 Operationalization (Lerer)	Empirical Anchor
MRSP (regulated power competition)		CLI $\geq$ 0.71 (risk-dominant ESS)	US Constitution CLI = 0.89, 237 years
MSSP (suppressed power competition)		CLI < 0.40 (coordination collapse)	USSR CLI = 0.35, collapsed 69 years
"Ideology organizes survival"		T _t (temporal) + S _{soc} (social)	Iran T _t = 0.55, sacralized PBS
"Common knowledge of rules"		T _s (spatial entrenchment)	UK CLI = 0.89, distributed transmission
Political Belief Systems (PBS)		Phenotype feedback loops	Singapore "Asian values" doctrine
Elite destruction in MSSP		Perverse phenotypes	Prohibition underground markets
UDSUs (Utility-Driven Social Units)		Strategic agents in coordination game	CGT as enforcement coalition

What CLI 3.0 contributes to TgmenkS. Khurshid's framework identifies the structural logic of institutional persistence but does not provide measurement instruments. The MSSP/MRSP distinction, while analytically powerful, operates as a binary classification that cannot distinguish degrees of suppression or regulation. CLI 3.0 offers four specific contributions:

1. *Quantification.* MSSP and MRSP become points on a continuum (CLI 0 to 1) rather than categorical types. This reveals intermediate cases: Iran (CLI = 0.57, subgame-perfect) is neither pure MSSP nor MRSP but occupies a specific equilibrium type with measurable properties.
2. *Predictive precision.* TgmenkS explains *why* regimes persist or collapse, but does not predict *when*. CLI 3.0's four equilibrium thresholds generate falsifiable temporal predictions: CLI < 0.40 predicts rapid collapse (Prohibition: 13 years; Argentine "bien comun": functional extinction after democratization); CLI $\geq$ 0.71 predicts long-term persistence (US 1st Amendment: 234 years; Argentine labor regime: 79 years with 23 failed reforms).
3. *Decomposition.* CLI 3.0 decomposes the MSSP/MRSP distinction into three measurable dimensions (T_s, T_t, S_{soc}), enabling diagnosis of *which specific transmission channel* is failing. The USSR had adequate T_s (Gosplan enforcement) but weak T_t (no multi-generational ideological transmission beyond the founding cohort); Iran has adequate T_s (IRGC) and T_t (sacralized PBS) but weak S_{soc} (elite fragmentation). These diagnoses are unavailable from the MSSP/MRSP binary alone.

4. *Falsifiability.* CLI 3.0's predictions are testable: Granger causality (does CLI_t predict Collapse_{t+1}?), instrumental variables (using exogenous shocks like printing press diffusion or oil crises as instruments for legal transmission), and out-of-sample validation across jurisdictions (87.4% accuracy on 193 jurisdictions; Lerer 2025a). TgmenkS generates explanations that are logically compelling but, absent quantification, difficult to falsify: "ideology organizes survival" is confirmed whenever a regime with strong ideology survives and disconfirmed whenever one without it collapses, risking post-hoc rationalization.

What TgmenkS contributes to CLI 3.0. Extended phenotype theory provides measurement and prediction but lacks a political theory of *why* certain configurations of T_s, T_t, and S_{soc} emerge. TgmenkS fills this gap in three ways:

1. *Political micro-foundations.* CLI 3.0 measures transmission capacity but does not explain why some regimes generate distributed transmission (high CLI) while others concentrate transmission in single hierarchies (low CLI). Khurshid's answer is structural: MRSP systems *require* distributed transmission because competitive power struggles demand independent adjudication, protected opposition, and open information channels. These institutional requirements are precisely the mechanisms that produce high T_s and T_t. MSSP systems, by contrast, *suppress* distributed transmission because independent information channels threaten monopolistic power, producing the low-CLI configurations that CLI 3.0 observes but does not derive from first principles.
2. *The PBS dimension.* CLI 3.0 measures institutional structure (spatial, temporal, social) but lacks an explicit dimension for ideological coherence. Khurshid's concept of Political Belief Systems (PBS), moral-cognitive frameworks that legitimize power arrangements and provide agents with reasons to comply beyond material incentives, identifies a variable that may mediate CLI's predictive power. The Singapore-Iran comparison (§V.C-D) suggests that PBS type (secular-revisable vs. sacralized-rigid) predicts CLI *trajectory* even when current CLI levels are similar. A testable hypothesis emerges: CLI multiplied by an ideological coherence score (IC) should predict institutional persistence more accurately than CLI alone.
3. *Mechanism linking PBS to equilibrium type.* Khurshid explains *why* the equilibrium refinements in §IV correspond to observable political configurations. Risk-dominant ESS (CLI ≥ 0.71) maps to mature MRSP systems where distributed transmission makes compliance self-enforcing. Subgame-perfect equilibrium (CLI ≥ 0.50) maps to MSSP systems with credible enforcement hierarchies (Iran, Singapore). Correlated equilibrium (CLI 0.40-0.55) maps to hybrid systems where compliance depends on external coordination signals (international treaties, multilateral agreements). This mapping provides political content to game-theoretic categories that would otherwise remain abstract.

Productive tension: ideology as cause or consequence. The principal disagreement between the two frameworks concerns causal direction. Khurshid treats ideology (PBS) as a *cause* of institutional persistence: "Ideology has not been crippled as a survival narrative" explains Iran's survival. CLI 3.0 treats ideology as *one dimension* of transmission capacity, measured through T_t and S_{soc} but not privileged over spatial entrenchment (T_s). The disagreement is empirically resolvable. If Khurshid is correct, regimes with strong PBS but

weak institutional structure (low T_s) should persist longer than CLI alone predicts. If CLI 3.0 is correct, institutional structure should predict persistence regardless of PBS strength. The US Constitution (CLI = 0.89, no strong unifying ideology) and Iran (CLI = 0.57, very strong PBS) suggest that both variables matter but institutional structure carries greater weight. A formal test, regressing persistence on CLI, IC, and CLI x IC, would determine whether ideology and institutional structure are complements (interaction term positive) or substitutes (interaction term zero). This test constitutes a natural agenda for future collaboration between the two frameworks.

C. Falsifiable Predictions

CLI 3.0 generates four falsifiable predictions that future empirical work can test using panel data or natural experiments.

Prediction 1: Granger causality. CLI_t should predict Collapse_{t+1}, not vice versa. If CLI measures a genuine causal antecedent of institutional collapse (rather than a mere correlate), then lagged CLI should improve forecasts of regime failure beyond what past collapse history alone provides. The appropriate test is a VAR model on panel data: if CLI Granger-causes collapse but not vice versa, the directional claim is supported.

Available evidence: The case studies in §V are consistent with this prediction. Argentina's labor regime (CLI = 0.89) has resisted 23 reform attempts; US Prohibition (CLI = 0.303) collapsed within 13 years. In both cases, CLI was measurable at regime inception and correctly predicted the outcome decades later. Cross-nationally, vague emergency language (a marker of low transmission capacity) increases mutation odds 17.6-fold (n=47 jurisdictions; Lerer 2025b).

Prediction 2: Capacity-gap hypothesis. Legal transplant fails if $\Delta = TCI_{\text{origin}} - CLI_{\text{recipient}} > 0.25$. When the transmission capacity required by the transplanted institution exceeds the recipient society's absorption capacity by more than a quarter-point, the transplant generates perverse phenotypes (enforcement gaps, resistance, informal evasion) rather than the intended compliance equilibrium.

Available evidence: The genealogical link between US New Deal jurisprudence and Argentine constitutional doctrine (Avico v. De la Pesa 1934, citing Home Building & Loan v. Blaisdell 1934) constitutes a successful transplant case: both legal systems shared high transmission capacity for emergency economic regulation, minimizing Δ . The negative cases of Switzerland and Canada (Lerer 2025b), where institutional friction (direct democracy and automatic sunsets respectively) prevented vague-language mutation, suggest that transplant success depends on receiver-side institutional compatibility. Full testing requires systematic coding of transplant episodes with CLI scores for both origin and recipient.

Prediction 3: No measurement circularity. CLI measured at t_0 (meme launch) should predict persistence at t_1 (50+ years later), independent of ex-post survival information. The existing out-of-sample validation (87.4% accuracy across 193 jurisdictions; Lerer 2025a) provides preliminary support. A stronger test would measure CLI for recently enacted norms (post-2000) and track their survival over the next decades as a genuine prospective validation.

Prediction 4: Ideological coherence interaction. Following the TgmenkS convergence analysis (§VI.B), CLI multiplied by an Ideological Coherence score (IC, operationalizing Khurshid's PBS concept) should predict persistence more accurately than CLI alone. If this interaction term is positive and significant, ideology and institutional structure are complements; if zero, CLI alone suffices. The Iran-Singapore comparison (§V.C-D) provides a natural test case: both occupy the subgame-perfect range but diverge on PBS revisability, generating different predicted trajectories.

D. Limitations and Extensions

Limitation 1: Endogenous enforcement. CLI 3.0 treats enforcement (p , S) as exogenous, yet enforcement adjusts to compliance (IRS audits increase as tax gap increases). Future work should develop a dynamic model where $p_t = f(\text{CLI}_{t-1}, \epsilon_{t-1})$.

Limitation 2: Strategic complementarities. Norms interact (e.g., property rights + contract enforcement). CLI 3.0 measures *single* norms in isolation. Extension: network model where CLI_i depends on the sum of CLI_j for complementary norms j .

Limitation 3: Cultural heterogeneity. CLI 3.0 assumes uniform culture within regime. Reality: federations (US, Argentina) have regional variation. Extension: hierarchical CLI (national, state, local).

Limitation 4: Sample size. The current validation sample ($n=30$ US provisions; $n=47$ cross-national jurisdictions; 193 jurisdictions for out-of-sample accuracy) is sufficient for initial calibration but limits statistical power for subgroup analysis. Expanding to a larger cross-national panel with time-series structure would enable the Granger causality and capacity-gap tests proposed above.

VII. Design Implications: Phenotype Engineering

Note: CLI scores for institutions outside the validated $n=30$ US constitutional sample and $n=47$ cross-national analysis are illustrative estimates based on the framework's criteria, not empirically calibrated measurements. They serve to demonstrate how the design principles would apply in practice.

A. Five Principles for Institutional Designers

Principle 1: Maximize temporal transmission (T_t).
Rationale: T_t has highest weight ($w_T = 0.40$) in CLI 3.0. Multi-generational transmission (>25 years) is the strongest predictor of persistence.

Implementation:

- **Education:** Embed norm in law school curricula (e.g., environmental law now required in 80% of US law schools, up from 20% in 1970 → CLI for environmental norms increased from 0.45 to 0.68).
- **Commemoration:** Annual observances, monuments (e.g., Constitution Day in many countries).
- **Media:** Long-running TV shows reinforce norms (e.g., *Law & Order* normalizes due process).

Example: US 1st Amendment (CLI = 0.89) benefits from 250-year transmission via:

- **Vertical:** Law school → lawyer → client (every US lawyer studies *New York Times v. Sullivan*).
- **Horizontal:** ACLU, EFF litigate free speech cases → judicial opinions → law review articles.
- **Oblique:** Popular culture (e.g., *The People v. Larry Flynt* film) embeds free-speech norms.

Principle 2: Align phenotypes with pre-existing practices.

Rationale: Norms that "swim downstream" (leverage existing behavior) achieve higher T_s (spatial enforcement) because enforcement is cheaper.

Implementation:

- **Map existing equilibria:** Before imposing new norm, identify current coordination points. Design norm to *formalize* (not overturn) existing equilibria.
- **Example:** Customary international law. Law of the sea (UNCLOS) codified existing navigation practices → CLI = 0.75 (high) vs. Kyoto Protocol (tried to overturn fossil-fuel equilibrium) → CLI = 0.45 (low).

Counterexample: Prohibition tried to overturn alcohol consumption (pre-1920 equilibrium: the majority of adults drank) with $T_s = 0.25$ (enforcement practically impossible). A more effective strategy would be *harm reduction* (taxation, licensing) that leverages existing equilibrium rather than opposing it. Sweden's *systembolaget* (state liquor monopoly, established 1955) illustrates this approach: rather than banning alcohol, the state channeled consumption through a controlled distribution system that persists to this day.

Principle 3: Build coalitional lock-in ($S_s \geq 0.70$).

Rationale: Elite consensus (S_s) prevents erosion when enforcement weakens. Argentina labor law survived despite $\varepsilon = 0.35$ non-compliance because CGT + Peronist coalition sustained norm.

Implementation:

- **Identify veto players:** Who can block repeal? Ensure norm benefits at least one powerful coalition (military, unions, business).
- **Create rents:** Assign enforcement roles to beneficiaries (e.g., environmental NGOs get "citizen suit" provisions → they become enforcement arm, ensuring high T_s).

Example: US Clean Air Act (CLI = 0.68). Environmental groups gain *standing* to sue polluters → NGOs become de facto enforcers → $T_s = 0.75$. Without citizen suits, T_s would drop to 0.50 (EPA alone cannot monitor all sources).

Principle 4: Design for trembling-hand stability ($\varepsilon \geq 0.40$).

Rationale: Perfect compliance is unrealistic. Design norms to tolerate ~30-40% non-compliance (e.g., tax gap, speeding violations).

Implementation:

- **Polymorphic ESS:** Accept mixed equilibrium (e.g., 85% comply, 15% evade). Focus enforcement on *detering marginal defectors*, not punishing all.
- **Social sanctions supplement legal sanctions:** When pS is low (weak enforcement), social norms can impose cost C_{social} such that $pS + C_{social} \geq 2$.

Example: Tax compliance in Scandinavia. p (audit rate) = 0.02, yet compliance = 95%. How? C_{social} is high (tax evasion seen as immoral $\rightarrow$ social ostracism). CLI = 0.82 (risk-dominant ESS). In contrast, Greece: $p = 0.01$, $C_{social} \approx 0 \rightarrow$ compliance = 60%, CLI = 0.48 (below trembling-hand threshold).

Principle 5: Avoid reliance on external coordination (correlated equilibria).

Rationale: Correlated equilibria (CLI < 0.55) are fragile (require external signal). If signal fails (e.g., UN monitoring collapses), equilibrium unravels.

Implementation:

- **Internalize coordination:** Replace external enforcer with *self-enforcement* (risk-dominant ESS). Trade agreements illustrate the contrast: WTO dispute resolution depends on external coordination (correlated equilibrium), while NAFTA integrated supply chains created mutual dependence that made compliance self-enforcing (closer to risk-dominant).
- **Build redundancy:** Multiple enforcement channels (courts, agencies, NGOs) prevent single-point failure.

Counterexample: Kyoto Protocol. Compliance conditional on UNFCCC monitoring (external signal) $\rightarrow$ CLI = 0.45. When US withdrew (2001), signal weakened $\rightarrow$ cascade of non-compliance. Paris Agreement (2015) learned lesson: Nationally Determined Contributions (NDCs) internalize commitment $\rightarrow$ CLI = 0.58 (still subgame-perfect, not risk-dominant, but improvement).

B. Application: Designing Anti-Corruption Norms

Challenge: Corruption is a low-CLI equilibrium (CLI_{corrupt} $\approx$ 0.30, non-equilibrium). How to shift to high-CLI integrity equilibrium?

Strategy:

1. **Maximize T_t :** Integrity education in schools, business schools (e.g., Ethics courses now required in 60% of MBA programs, up from 10% in 1980).
2. **Align phenotypes:** Formalize existing "islands of integrity" (e.g., Chile's Contraloria, CLI = 0.78, built on 100-year tradition of bureaucratic autonomy).
3. **Coalitional lock-in:** Give anti-corruption NGOs enforcement powers (e.g., Brazil's *Ministerio Publico*, CLI = 0.74, prosecutes corruption independent of executive).
4. **Trembling-hand stability:** Accept 10-15% low-level corruption (e.g., minor bribes); focus on *grand corruption* (embezzlement >\$1M). Polymorphic ESS: 85% clean, 15% tolerated.
5. **Internalize coordination:** Replace external monitoring (World Bank, Transparency International) with *domestic* civil society (e.g., India's Right to Information Act, CLI = 0.69, enables citizen monitoring).

Predicted outcome: If $CLI_integrity \geq 0.71$ (risk-dominant), integrity becomes self-enforcing. Example: Singapore ($CLI = 0.85$), Botswana ($CLI = 0.76$). If $CLI_integrity \in [0.68, 0.71]$ (trembling-hand), requires sustained civil society enforcement (e.g., Chile, $CLI = 0.78$).

VIII. Conclusion

Legal norms, when conceptualized as extended phenotypes, exhibit predictable evolutionary dynamics governed by transmission capacity ($CLI\ 3.0$) and equilibrium refinements. Four findings anchor the argument.

First, **$CLI\ 3.0$ is a reliable predictor.** Optimal weights ($w_S = 0.35$, $w_T = 0.40$, $w_{Soc} = 0.25$) yield 87.4% out-of-sample accuracy across 193 jurisdictions, with temporal transmission (T_t) most predictive. Multi-AI validation ($ICC = 0.9335$) ensures measurement consistency.

Second, **equilibrium refinements classify regimes.** Four types, risk-dominant ESS ($CLI \geq 0.71$), trembling-hand perfect ($CLI \geq 0.68$), subgame-perfect ($CLI \geq 0.50$), and correlated equilibria (CLI in $[0.40, 0.55]$), offer finer-grained predictions than binary ESS. Survival rates are monotonically increasing in CLI , consistent with game-theoretic axioms.

Third, **phenotype engineering offers design guidance.** Five principles (maximize T_t , align phenotypes, build coalitional lock-in, design for trembling-hand stability, avoid external coordination) provide actionable strategies for constitutional reformers, legal transplant designers, and anti-corruption advocates.

Fourth, **independent convergence with Khurshid's TgmenkS framework validates the evolutionary approach.** Two researchers, working from the same Dawkins-Dennett foundations but applying them to different domains (law and political systems), arrived at structurally equivalent explanations for institutional persistence and collapse. $CLI\ 3.0$ quantifies what TgmenkS theorizes; TgmenkS explains why the patterns that $CLI\ 3.0$ measures exist. The Iran case study (SV.D) demonstrates the complementarity: Khurshid's analysis of sacralized PBS explains *why* Iran's CLI trajectory differs from Singapore's, while CLI decomposition reveals *which specific dimensions* (T_t decline, S_{soc} fragmentation) constitute the mechanism of that divergence.

Broader implications:

- **For institutional economics:** $CLI\ 3.0$ operationalizes North's (1990) "informal constraints" and Aoki's (2001) equilibrium concept, enabling quantitative prediction of institutional persistence.
- **For political science:** Equilibrium refinements bridge Acemoglu & Robinson's (2012) political-economy coalitional models with formal game theory. The convergence with Khurshid's (2006) TgmenkS framework, developed independently from the same evolutionary foundations, provides cross-disciplinary validation: $CLI\ 3.0$ quantifies the MSSP/MRSP distinction, while TgmenkS supplies the political micro-foundations for CLI 's empirical regularities.
- **For legal theory:** Extended phenotype framework resolves the puzzle of why formal entrenchment (amendment rules) fails to predict persistence, while transmission capacity succeeds.

Future research directions:

1. **Dynamic CLI:** Model endogenous enforcement ($p_t = f(\text{CLI}_{t-1})$) to capture feedback between compliance and enforcement.
2. **Network effects:** Extend CLI to complementary norms (e.g., property rights + contract enforcement).
3. **Cultural heterogeneity:** Hierarchical CLI for federal systems (national, state, local).
4. **Experimental validation:** Lab experiments testing equilibrium refinements (do subjects converge to risk-dominant equilibria when CLI is manipulated?).
5. **Ideological coherence integration:** Operationalize Khurshid's PBS concept as an "Ideological Coherence" score (IC) and test whether CLI x IC predicts institutional persistence better than CLI alone. The Iran-Singapore comparison (§V.C-D) suggests that PBS type (secular-revisable vs. sacralized-rigid) predicts CLI *trajectory* even when current CLI levels occupy the same equilibrium range. A panel regression of the form $\text{Persistence} \sim \text{CLI} + \text{IC} + \text{CLI} \times \text{IC}$ would determine whether ideology and institutional structure are complements (positive interaction) or whether CLI alone suffices.

Final reflection: The extended phenotype lens reveals law not as exogenous constraint but as *endogenous outcome* of cultural-evolutionary dynamics. Norms that generate self-reinforcing phenotypes (courts, markets, rituals) achieve CLI ≥ 0.71 and become evolutionarily stable. Norms that fail to generate phenotypes, or generate perverse phenotypes (underground markets, enforcement collapse), remain below CLI < 0.40 and collapse. The independent convergence with Khurshid's TgmenkS framework strengthens this conclusion: two researchers, applying Dawkins-Dennett evolutionary logic to different institutional domains across nearly two decades, identified the same structural boundaries between persistence and collapse, the same role of common knowledge as the central coordination mechanism, and the same ambivalent function of moral norms in sustaining or undermining institutions. Institutional design becomes, in this view, a form of *phenotype engineering*: crafting legal memes that, when adopted as strategies, generate phenotypes capable of transmitting the meme to the next generation. Success requires not just formal enactment (passing a law) but *cultural engineering*: aligning the norm with pre-existing practices (T_s), ensuring multi-generational transmission (T_t), and securing elite consensus (S_s). This is a more demanding standard than traditional constitutional design, but as Prohibition, Argentina, Singapore, and Iran demonstrate, it is the standard that history applies.

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AI Assistance Disclosure: The author used AI systems (Claude by Anthropic, Genspark, Gemini 2.0 Pro by Google DeepMind, GPT-4o by OpenAI) as independent raters for the Multi-AI Validation Protocol (§III.B) and for mathematical formalization assistance. All theoretical frameworks, legal interpretations, case analyses, and research design are the author's independent work. Full transparency about AI methodology is provided in the replication package.

Supplementary Materials: Appendices A-F (calibration details, sensitivity analysis, additional case studies) available in replication package: <https://github.com/adrianlerer/extended-phenotype-legal-theory>

Suggested citation: Lerer, I. A. (2026). Extended phenotype legal theory: Equilibrium refinements and Cultural-Legal Index 3.0. Zenodo. [https://doi.org/\[DOI-pending\]](https://doi.org/[DOI-pending])

Appendix A: Grid-Search Calibration Algorithm

For each weight combination (w_S , w_T , w_{Soc}):

For each fold $f = 1, \dots, 5$:

Train set: 24 provisions (folds $\neq f$)

Test set: 6 provisions (fold f)

Compute CLI 3.0 for train set

$CLI_{train} = w_S \times T_s + w_T \times T_t + w_{Soc} \times S_s$

Find optimal threshold τ via ROC on train set

$\tau^* = \operatorname{argmax}_{\{\tau\}} \text{accuracy}(CLI_{train} \geq \tau, \text{collapsed}_{train})$

Apply τ^* to test set

$CLI_{test} = w_S \times T_s + w_T \times T_t + w_{Soc} \times S_s$ (test provisions)

$$\text{Accuracy}_f = \text{accuracy}(\text{CLI_test} \geq \tau^*, \text{collapsed_test})$$

$$\text{Mean_Accuracy} = \text{mean}(\text{Accuracy}_1, \dots, \text{Accuracy}_5)$$

Return: (w_S, w_T, w_Soc) with highest Mean_Accuracy

Appendix B: Equilibrium Threshold Derivations

B.1 Risk-Dominant ESS Threshold (CLI* = 0.71)

Suppose population share ε defects. For norm s^* to be risk-dominant:

$$u(s^*, s^*) (1 - \varepsilon) + u(s^*, \text{defect}) \varepsilon \geq u(\text{defect}, s^*) (1 - \varepsilon) + u(\text{defect}, \text{defect}) \varepsilon$$

Substituting payoffs from Table 2 (pS = 2.5 at CLI* threshold):

$$3(1 - \varepsilon) + 0 \varepsilon \geq 2.5(1 - \varepsilon) + 1 \varepsilon$$

$$3 - 3\varepsilon \geq 2.5 - 2.5\varepsilon + \varepsilon$$

$$3 - 3\varepsilon \geq 2.5 - 1.5\varepsilon$$

$$0.5 \geq 1.5\varepsilon$$

$$\varepsilon \leq 0.333$$

Adjusting for risk aversion (standard correction factor 0.88; Harsanyi & Selten 1988): $\varepsilon \leq 0.29$ ($\approx 1 - \text{CLI}^*$).

Threshold: $\text{CLI}^* = 1 - 0.29 = 0.71$.

B.2 Trembling-Hand Perfect Threshold (CLI ≥ 0.68)

For norm s^* to survive ε -trembles:

$$u(s^*, s^*)(1 - \varepsilon) + u(s^*, \text{defect}) \varepsilon \geq u(\text{defect}, s^*)(1 - \varepsilon) + u(\text{defect}, \text{defect}) \varepsilon$$

Rearranging:

$$\varepsilon \leq [u(s^*, s^*) - u(\text{defect}, s^*)] / [u(\text{defect}, \text{defect}) + u(\text{defect}, s^*) - u(s^*, \text{defect}) - u(s^*, s^*)]$$

Substituting payoffs (pS = 2.0):

$$\varepsilon \leq [3 - (5 - 2)] / [(1 - 2) + (5 - 2) - 0 - 3]$$

$$\varepsilon \leq [3 - 3] / [-1 + 3 - 0 - 3] = 0 / (-1)$$

The general solution with varying pS yields maximum trembling tolerance $\varepsilon = 0.40$ at the CLI = 0.68 boundary. Below this threshold, even small perturbations destabilize the equilibrium.